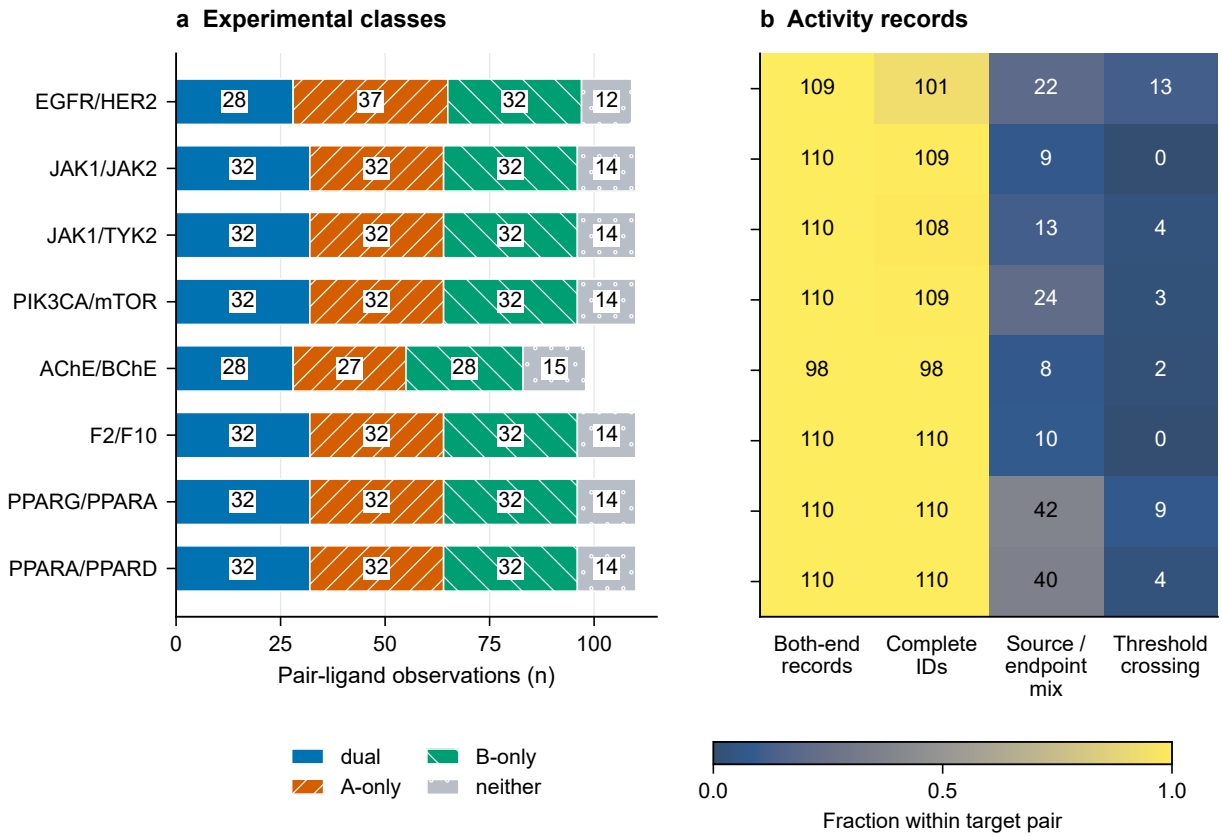


Fig. 1 Experimental classes and activity-source coverage. (a) Numbers of dual, A-only, B-only and neither observations in each target pair. (b) Counts with activity records on both targets, complete document identifiers, mixed sources or endpoints, and threshold-discordant repeated measurements. Cell color gives the fraction within each pair; labels give counts. Coverage flags overlap. Complete identifiers require DOI, PubMed or patent identifiers for all tied activity maxima at both endpoints. The 867 observations represent 845 distinct ligand entities

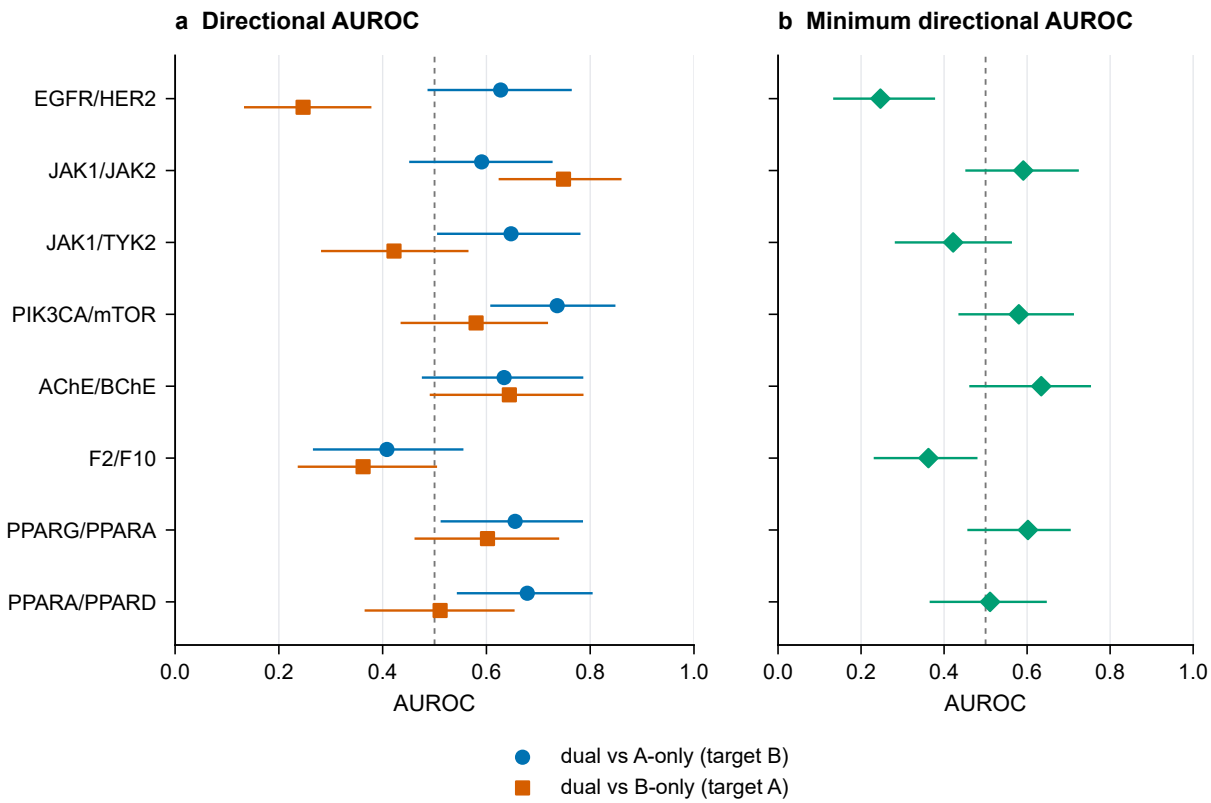


Fig. 2 Vina directional discrimination and minimum AUROC. Points and horizontal bars show estimates and 95% class-stratified percentile bootstrap intervals. (a) Dual versus A-only uses target B; dual versus B-only uses target A. (b) Minimum AUROC is the smaller directional AUROC, calculated within each resample. The vertical line marks AUROC 0.5 as a reference. The summary interval is not a simultaneous test that both directions exceed or fall below 0.5

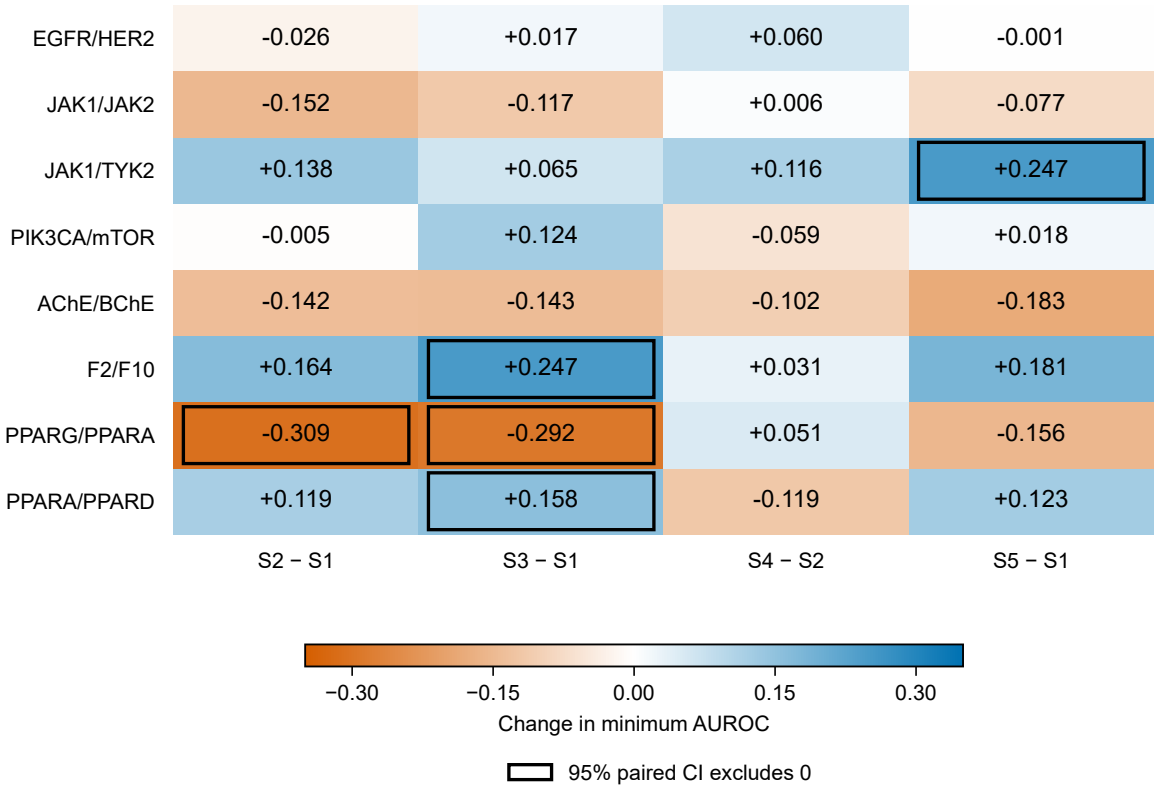


Fig. 3 Paired changes in minimum AUROC across target pairs and scoring workflows. Cells show signed three-decimal differences on a common symmetric color scale centered at zero: S2-S1, fixed-pose GNINA versus Vina; S3-S1, fixed-pose RTMScore versus Vina; S4-S2, saved-pose reranking; and S5-S1, independent GNINA docking versus Vina. Each contrast uses its own complete-case paired population. A black border indicates that the 95% paired bootstrap interval excludes zero; an unbordered cell does not imply equivalence. Border status does not encode interval width and is not adjusted for multiple comparisons. Full estimates and intervals are retained in Figure S4 and Table S10B. No across-pair pooled estimate is calculated

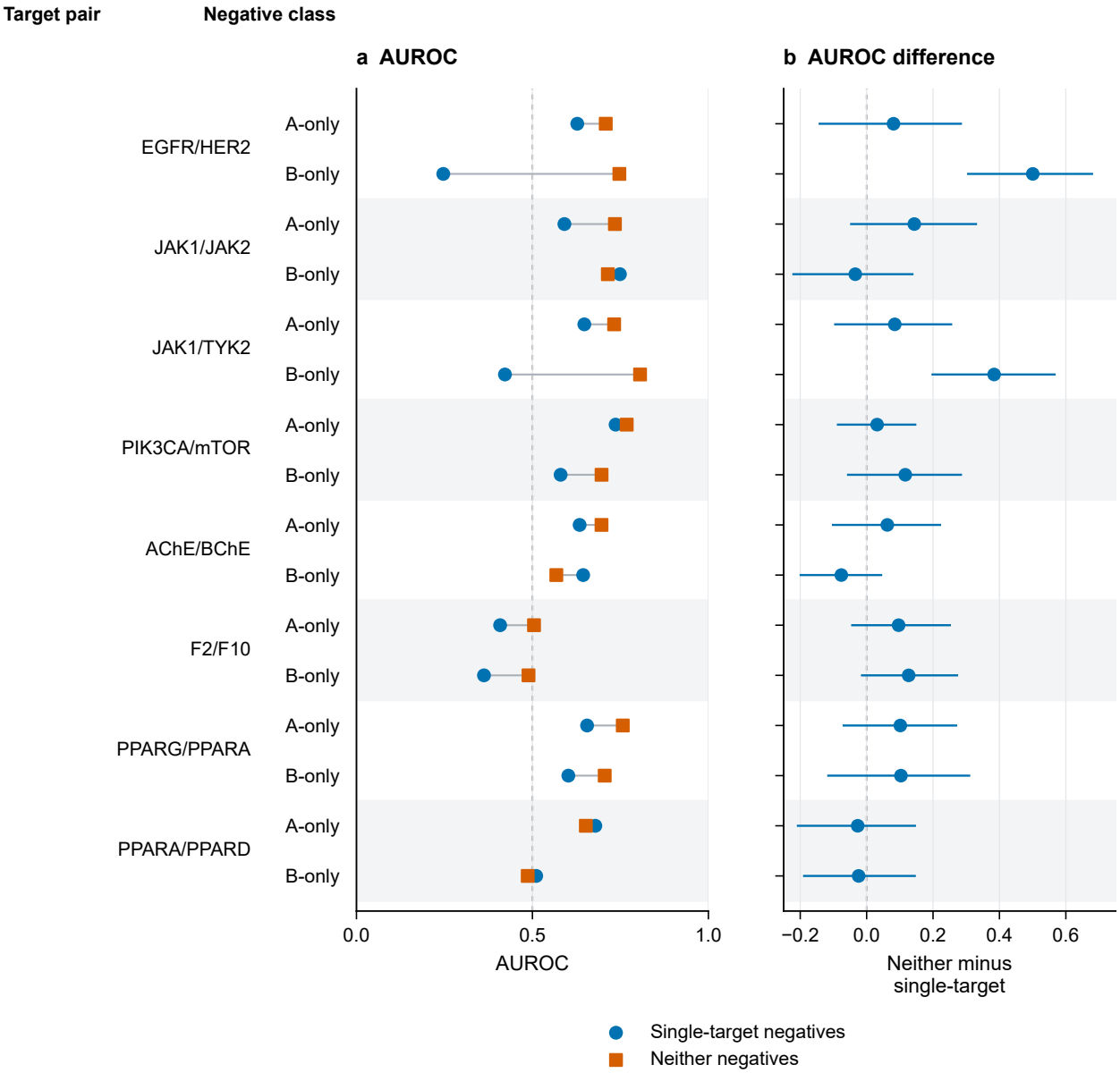


Fig. 4 Effect of negative-class composition. (a) AUROC using single-target or neither negatives; the dual members and score target are held fixed within each comparison. (b) Neither minus single-target AUROC with 95% bootstrap confidence intervals. The two negative classes are resampled separately with shared dual resamples. All scores are Vina S1. Row labels identify the single-target negative class: A-only uses target B scores; B-only uses target A scores

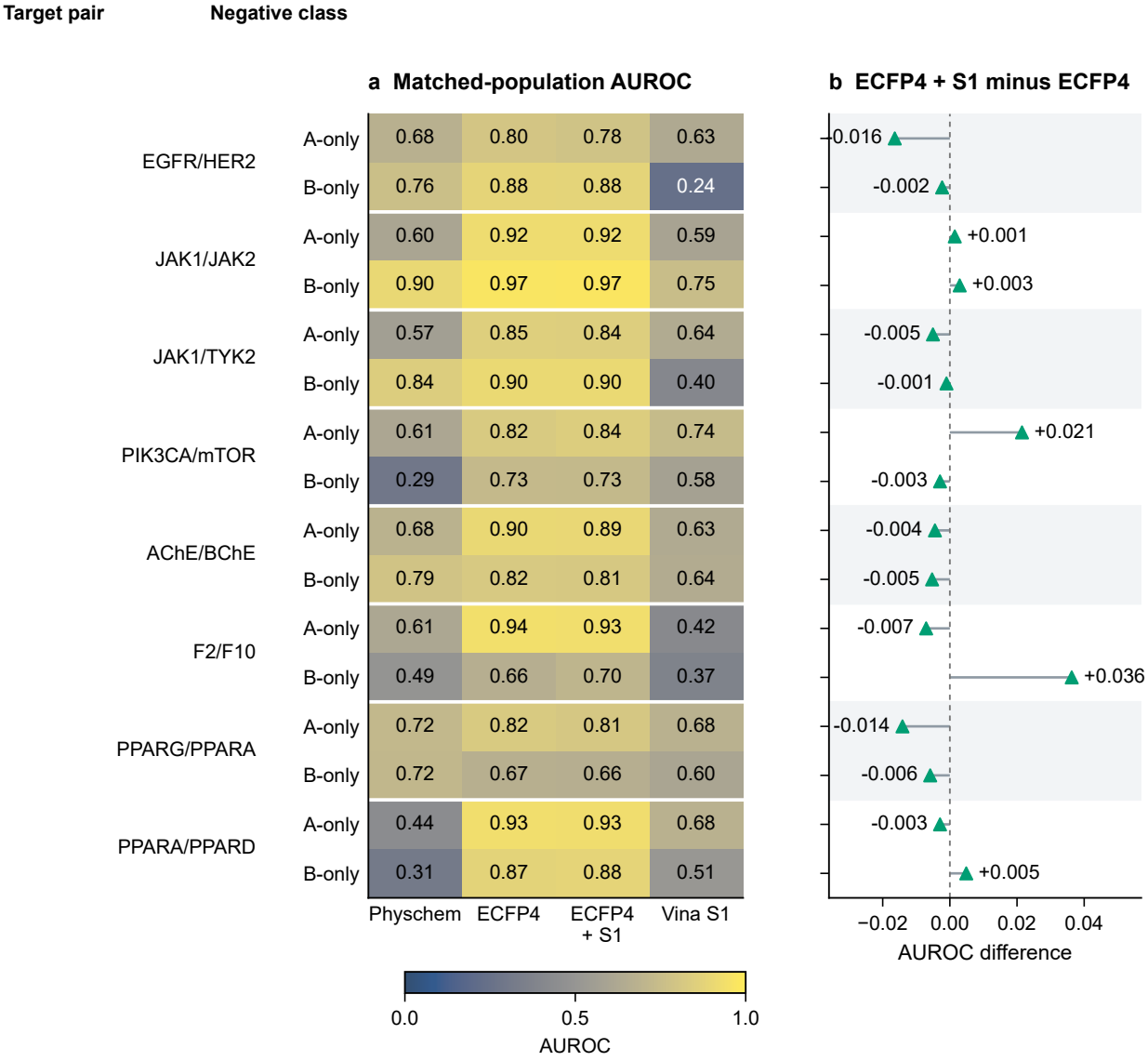


Fig. 5 Ligand chemistry and the incremental contribution of Vina. (a) Matched-population AUROC for physicochemical descriptor (Physchem), ECFP4, ECFP4-plus-S1 models and direct Vina S1 scores. Cells show two-decimal estimates on a common 0-1 color scale. The three learned models use five-fold out-of-fold predictions. (b) AUROC(ECFP4 plus S1) minus AUROC(ECFP4); stems start at zero and labels show signed three-decimal differences. The two panels use different numeric scales. Confidence intervals were not calculated for these chemistry comparisons. Dual is the positive class; A-only and B-only rows use target B and A scores, respectively

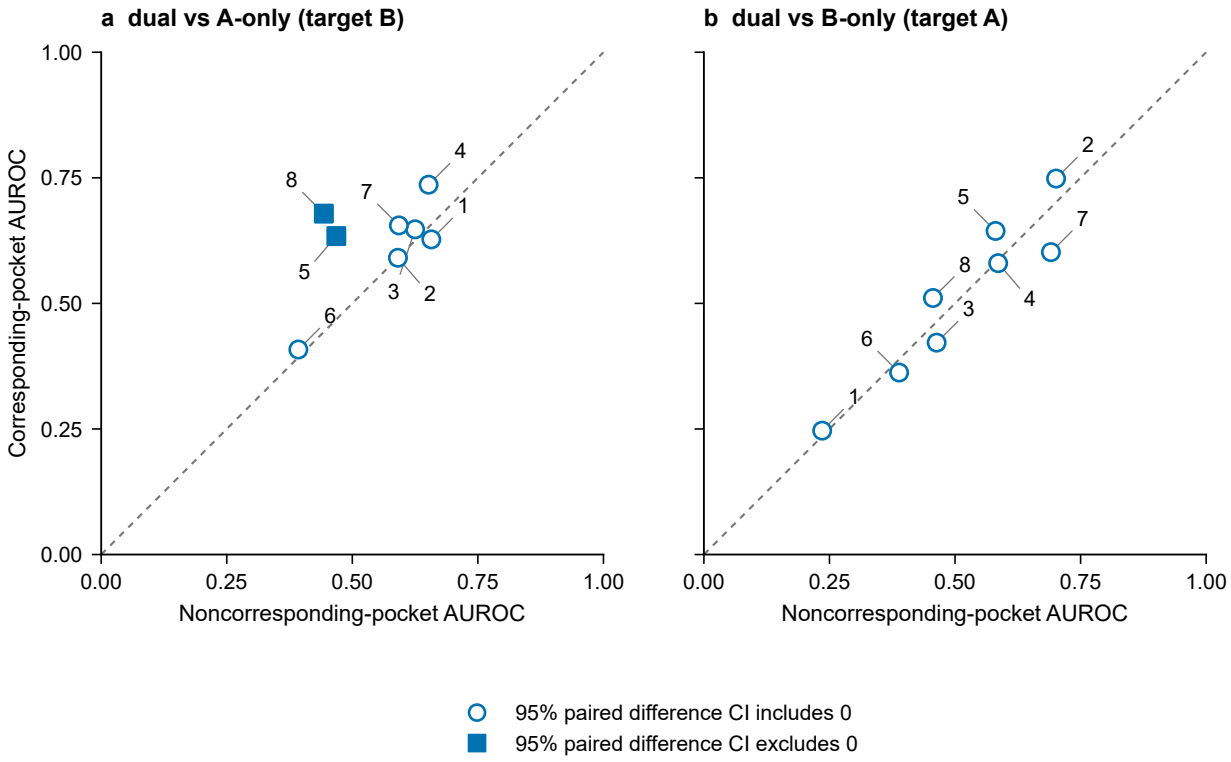


Fig. 6 Corresponding versus noncorresponding pocket AUROC. Each numbered point identifies a target pair on the same complete-case directional population. (a) Dual versus A-only, with target B as the corresponding pocket. (b) Dual versus B-only, with target A as the corresponding pocket. The dashed identity line marks equal AUROC; points above it have higher corresponding-pocket AUROC. Filled squares indicate that the 95% paired bootstrap interval for corresponding minus noncorresponding AUROC excludes zero; open circles indicate that it includes zero. Symbols summarize interval status, not a simultaneous confidence region or a multiplicity-adjusted test. All scores are Vina S1. Full original estimates, paired differences and interval widths are retained in Figure S8 and Table S5B. Target-pair codes: 1, EGFR/HER2; 2, JAK1/JAK2; 3, JAK1/TYK2; 4, PIK3CA/mTOR; 5, AChE/BChE; 6, F2/F10; 7, PPARG/PPARA; 8, PPARA/PPARD

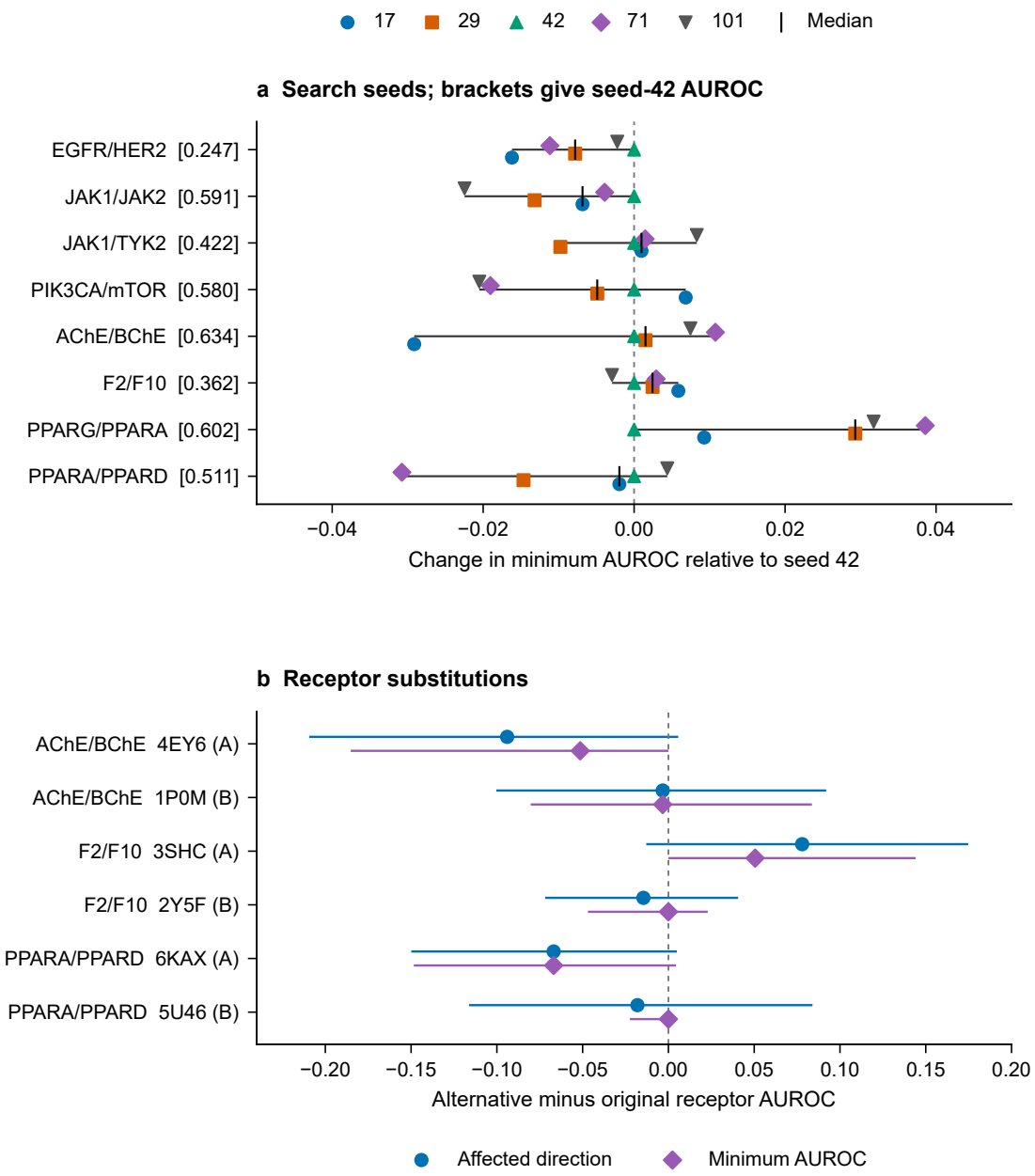


Fig. 7 Sensitivity to search seeds and receptor structures. (a) Seed-specific Vina minimum AUROC minus the corresponding seed-42 value. Brackets in row labels give the absolute seed-42 AUROC. Segments show the observed minimum-to-maximum seed range after subtracting seed 42; black ticks show the median change. These ranges are not confidence intervals. All five seed results are retained; each uses its valid members. AChE/BChE seed 17 has one fewer A-only ligand than seed 42, so the displayed difference includes that membership change. Absolute seed-specific estimates are shown in Figure S9. (b) Paired changes in affected-direction AUROC and minimum AUROC for six single-target receptor replacements, with 95% paired bootstrap intervals. Parentheses identify replaced side A or B; the other receptor is held fixed

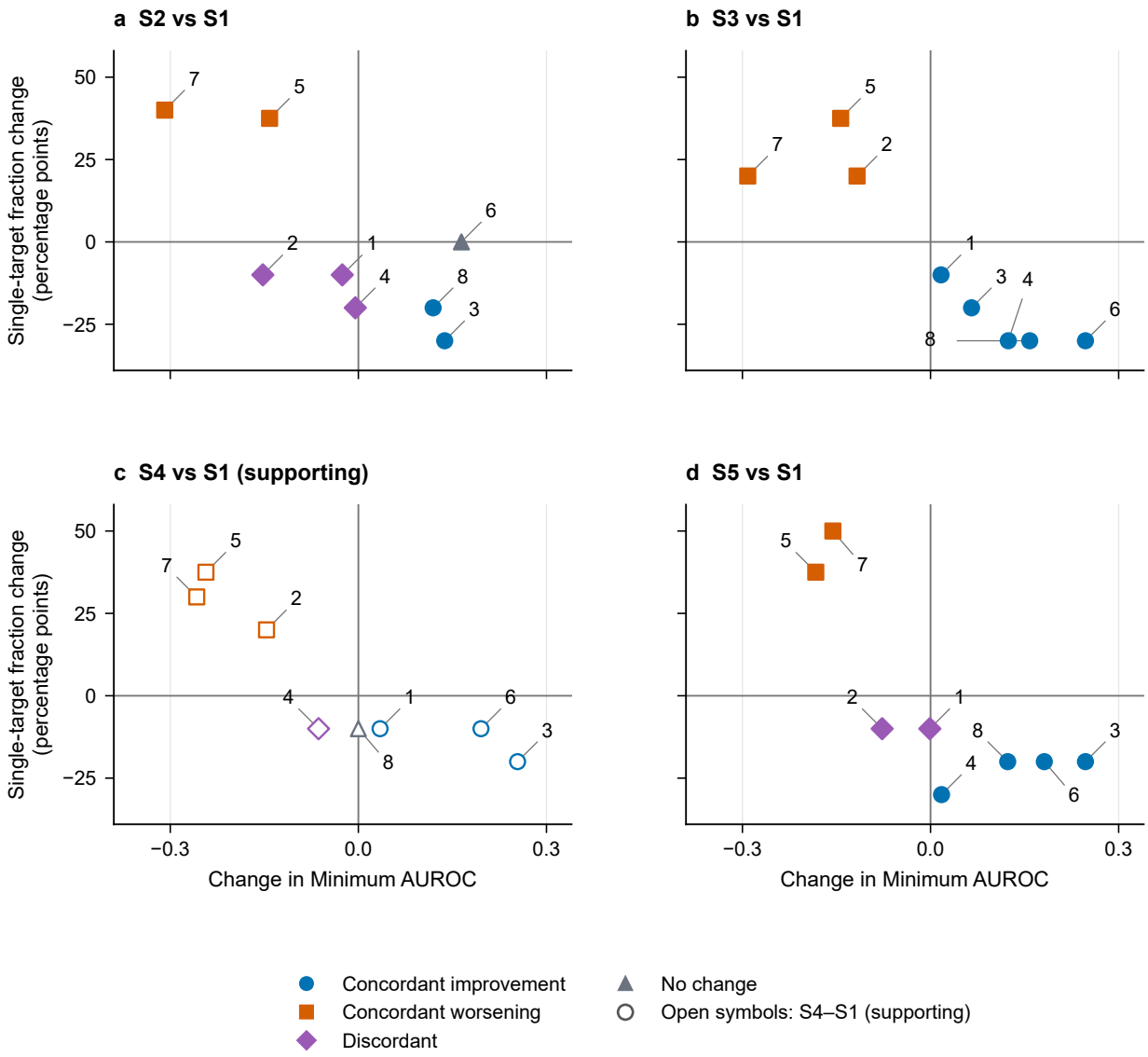


Fig. 8 Relationship between directional discrimination and candidate-list composition. Each point is one target pair. Both axes show the new scheme minus S1. The horizontal axis is the paired change in minimum AUROC; the vertical axis is the change in single-target-selective fraction among the top 10% of joint ranks, expressed in percentage points. Ranking and directional changes use their respective complete-case populations, as defined in Methods. Colors and symbols identify the four change categories. Open symbols identify the supporting S4-S1 comparison. Matched top-list counts are shown in Table 2 and Figure S12. Target-pair codes: 1, EGFR/HER2; 2, JAK1/JAK2; 3, JAK1/TYK2; 4, PIK3CA/mTOR; 5, AChE/BChE; 6, F2/F10; 7, PPARG/PPARA; 8, PPARA/PPARD

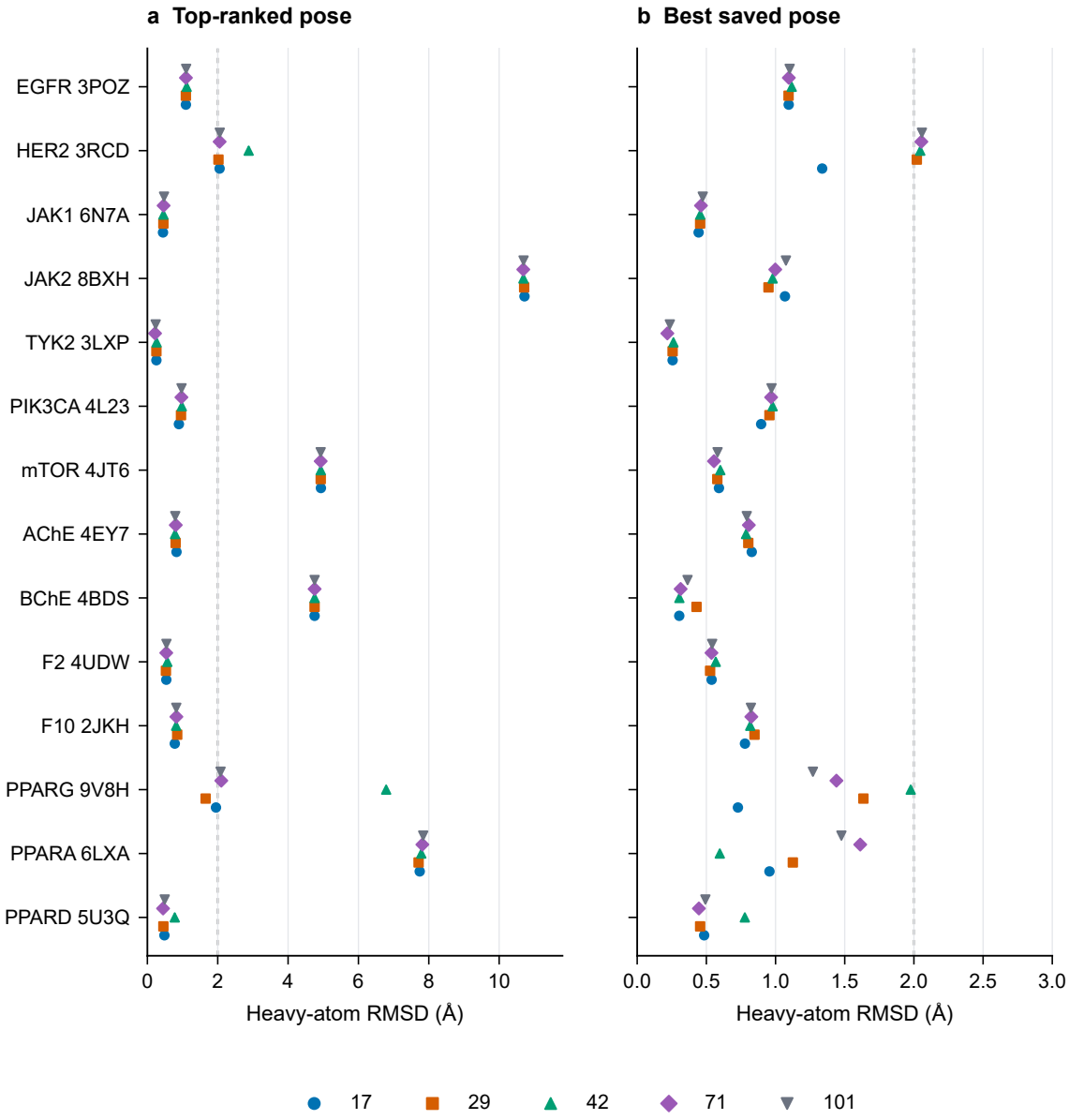


Fig. S1 Cocrystal-ligand redocking for 14 primary receptors and five search seeds. (a) Top-ranked-pose RMSD. (b) Minimum RMSD among all saved poses. RMSDs are symmetry-aware heavy-atom distances computed in the receptor frame. The dashed line marks 2 angstrom. Each point is a seed result; no coordinate or RMSD is recomputed for this figure. Axes differ between panels to preserve visibility of large top-ranked-pose errors. Seed markers have small fixed vertical offsets for visibility; offsets do not change RMSD values

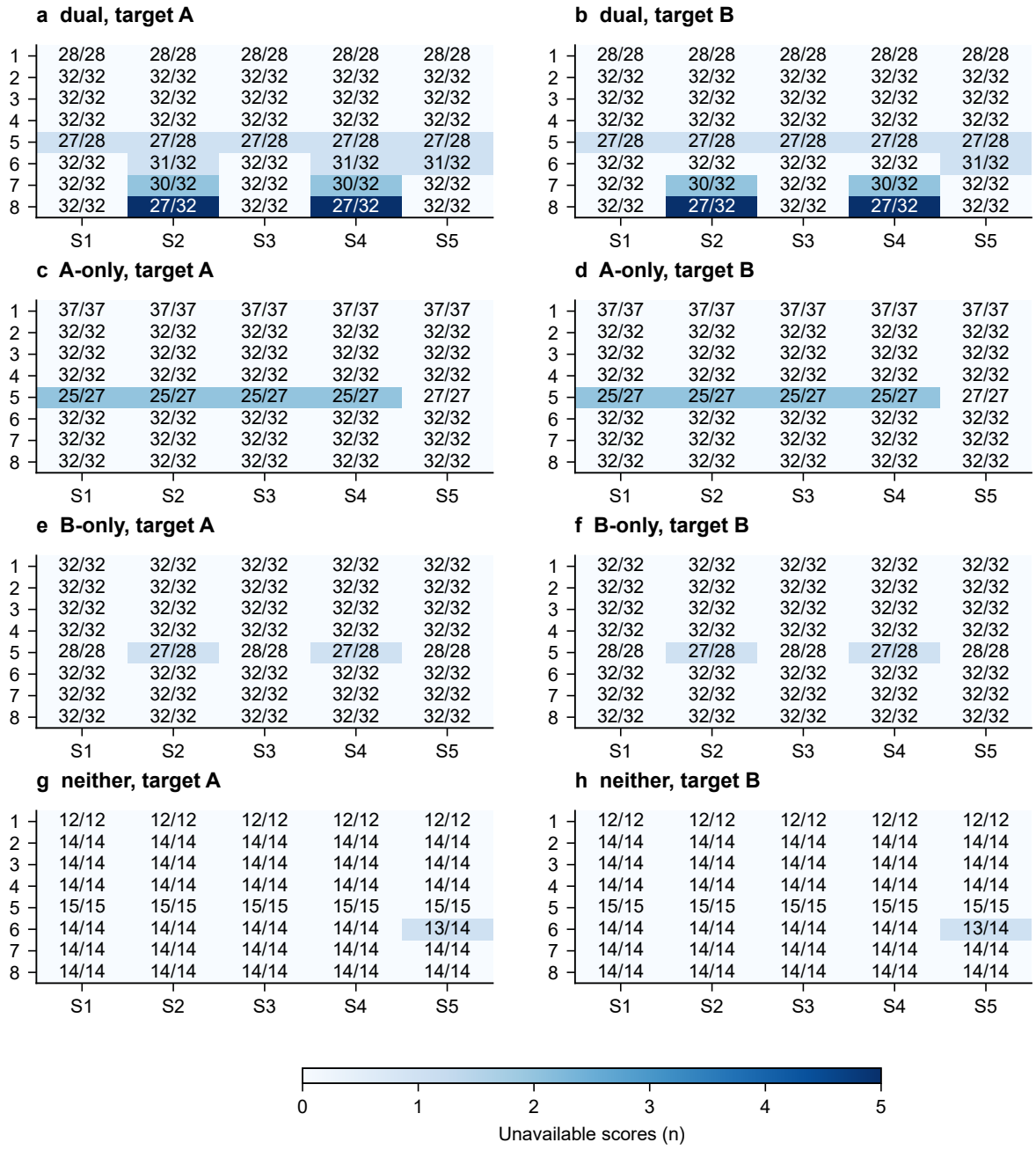


Fig. S2 Availability of scores by experimental class, target and scoring scheme. Each cell gives valid/expected counts; its color gives the number of unavailable or invalid scores on a common 0-5 scale, with white indicating complete availability. Missing and invalid scores retain their original status, and are not imputed. The S1-S5 order follows the Methods scoring definitions. Target-pair codes: 1, EGFR/HER2; 2, JAK1/JAK2; 3, JAK1/TYK2; 4, PIK3CA/mTOR; 5, AChE/BChE; 6, F2/F10; 7, PPARG/PPARA; 8, PPARA/PPARD

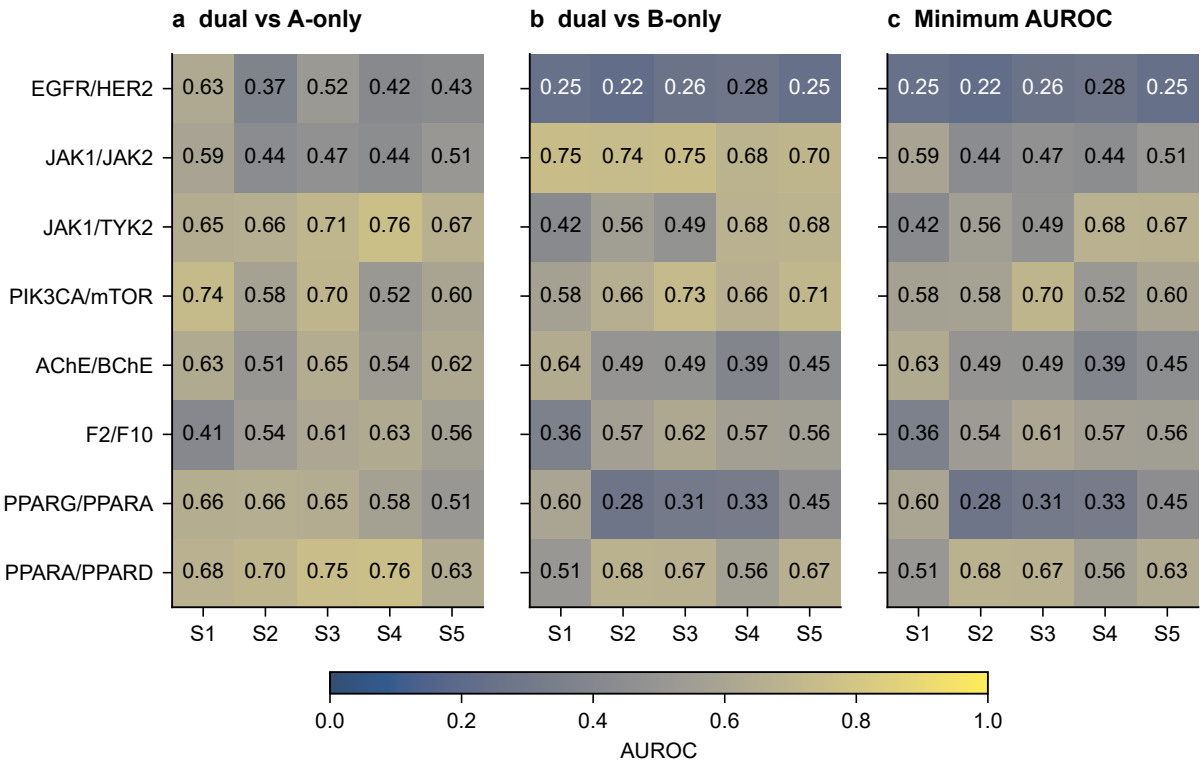


Fig. S3 Directional AUROC and minimum AUROC for all five scoring schemes. Cell annotations show estimates rounded to two decimals; colors use a common 0-1 scale. Intervals and sample sizes are given in Table S10A. The heat map does not replace paired method comparisons, which use their own shared members

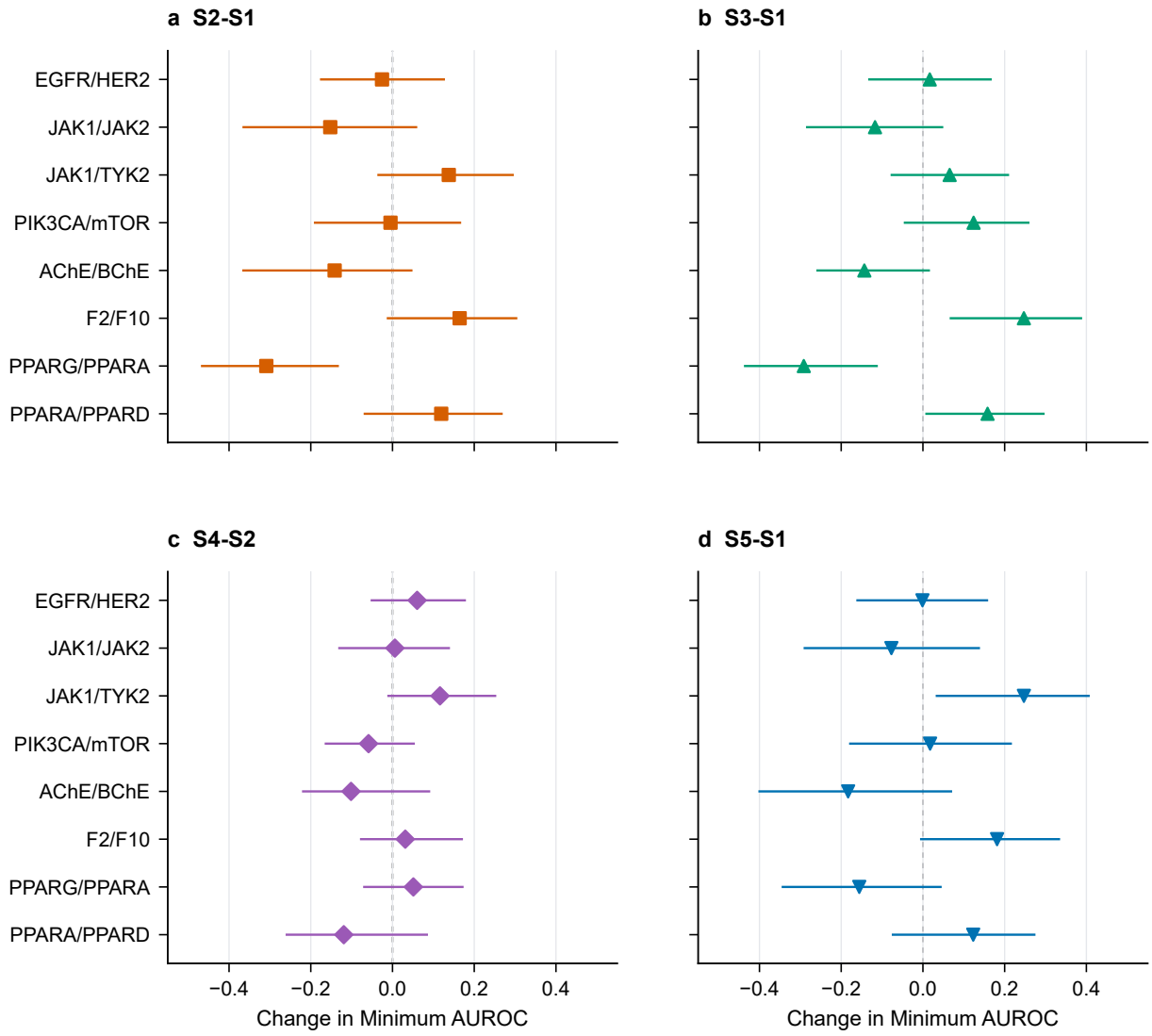


Fig. S4 Full paired scoring-workflow intervals underlying Figure 3. Paired changes in minimum AUROC between scoring workflows. (a) S2-S1, fixed-pose GNINA versus Vina. (b) S3-S1, fixed-pose RTMScore versus Vina. (c) S4-S2, selection of the highest CNNscore among saved Vina poses versus the original mode-1 pose. (d) S5-S1, independent GNINA docking versus Vina. Every contrast uses its own complete-case paired members. Bars are 95% paired bootstrap intervals; no across-pair pooled estimate is calculated

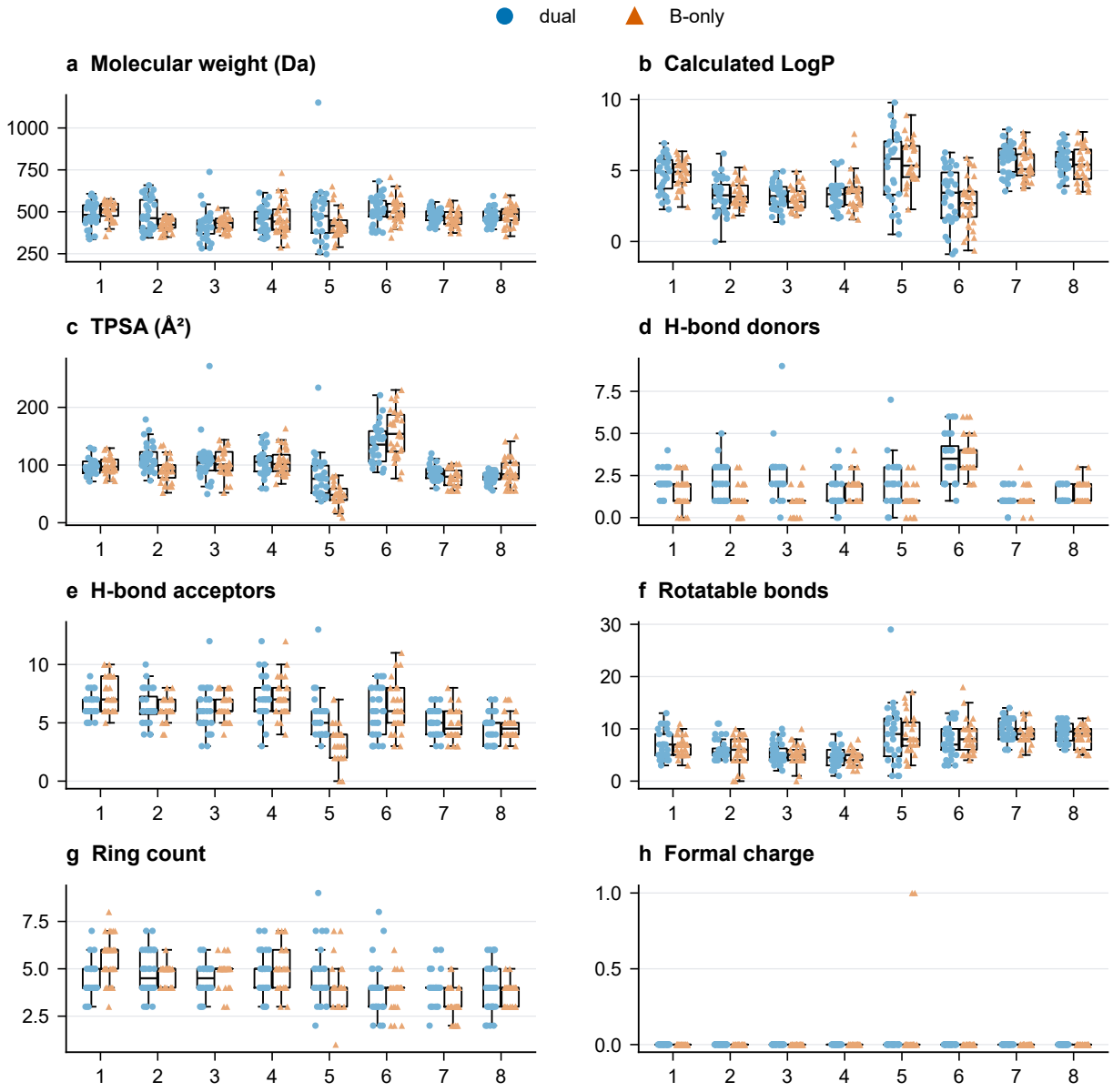


Fig. S5 Physicochemical descriptor distributions for dual and B-only ligands. Points are labeled-population observations before score and fold filtering. Boxes show the median and interquartile range, with whiskers to the most extreme values within 1.5 IQR. All observations are retained as points, including those beyond whiskers. These distributions are descriptive and do not imply independently validated classification. Target-pair codes: 1, EGFR/HER2; 2, JAK1/JAK2; 3, JAK1/TYK2; 4, PIK3CA/mTOR; 5, AChE/BChE; 6, F2/F10; 7, PPARG/PPARA; 8, PPARA/PPARD

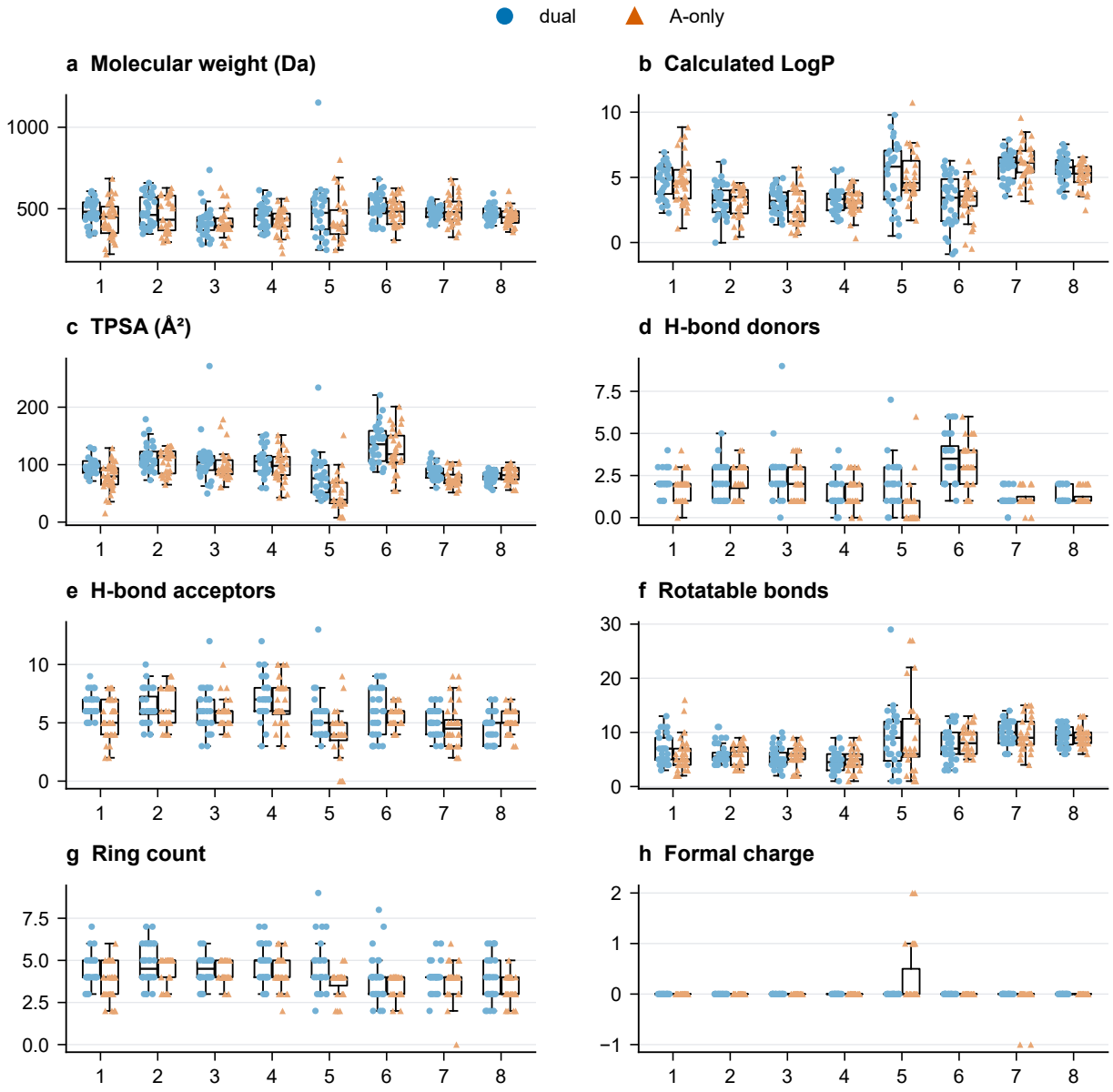


Fig. S6 Physicochemical descriptor distributions for dual and A-only ligands. Points are labeled-population observations before score and fold filtering. Boxes show the median and interquartile range, with whiskers to the most extreme values within 1.5 IQR. All observations are retained as points, including those beyond whiskers. These distributions are descriptive and do not imply independently validated classification. Target-pair codes: 1, EGFR/HER2; 2, JAK1/JAK2; 3, JAK1/TYK2; 4, PIK3CA/mTOR; 5, AChE/BChE; 6, F2/F10; 7, PPARG/PPARA; 8, PPARA/PPARD

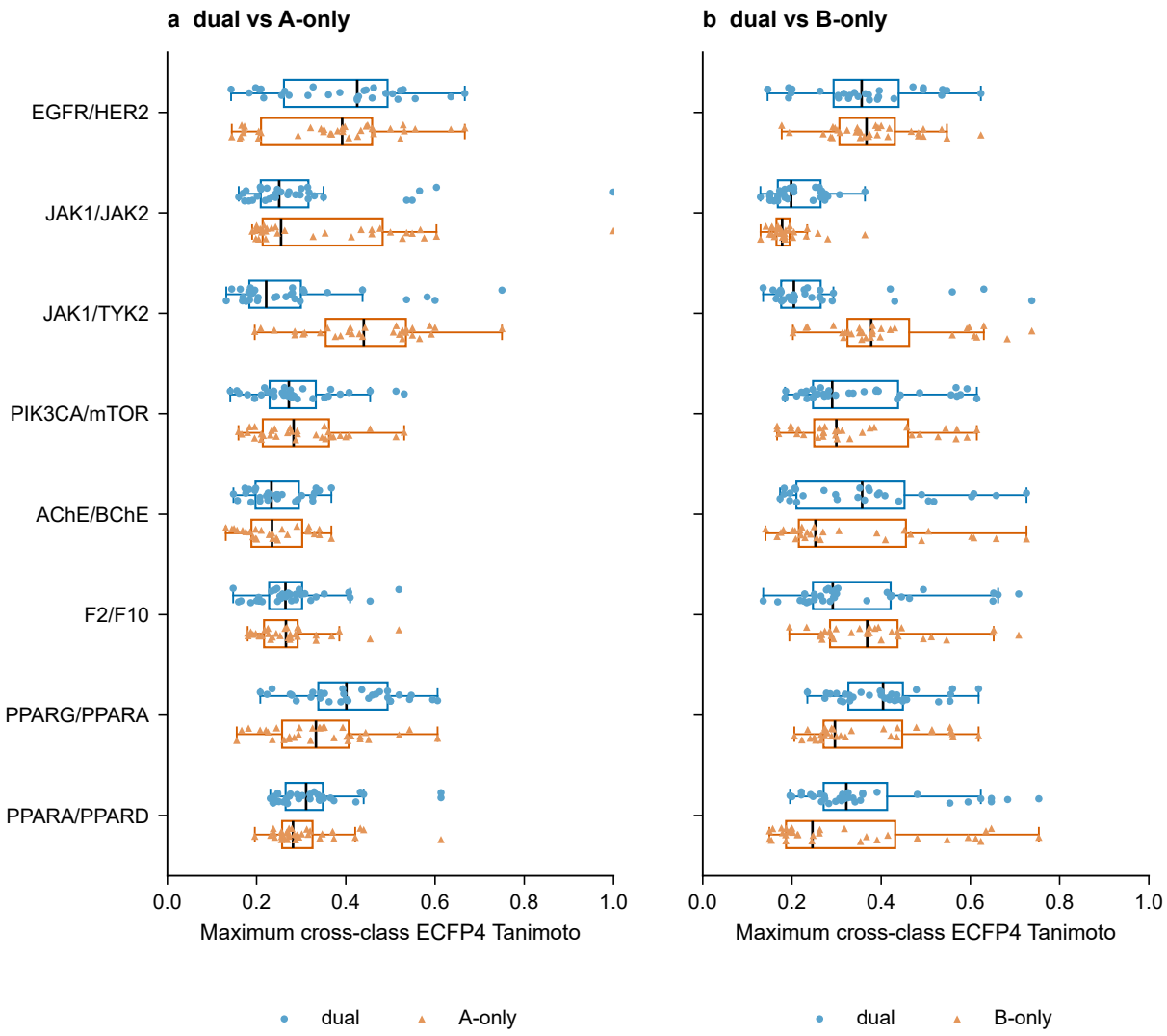


Fig. S7 Structural similarity between the two experimental classes within each directional task. Each point is the maximum ECFP4 Tanimoto similarity to any ligand in the other class of the same target pair and direction. Fingerprints use radius 2, 2048 bits, useChirality=False and useFeatures=False. Dual and single-target classes have separate boxes and vertically separated points. Boxes show the median and interquartile range; whiskers extend to the most extreme values within 1.5 IQR. All observations, including those beyond the whiskers, are shown. These are descriptive class-conditioned nearest-neighbor similarities, not independent replicate measurements or a cross-validation leakage test

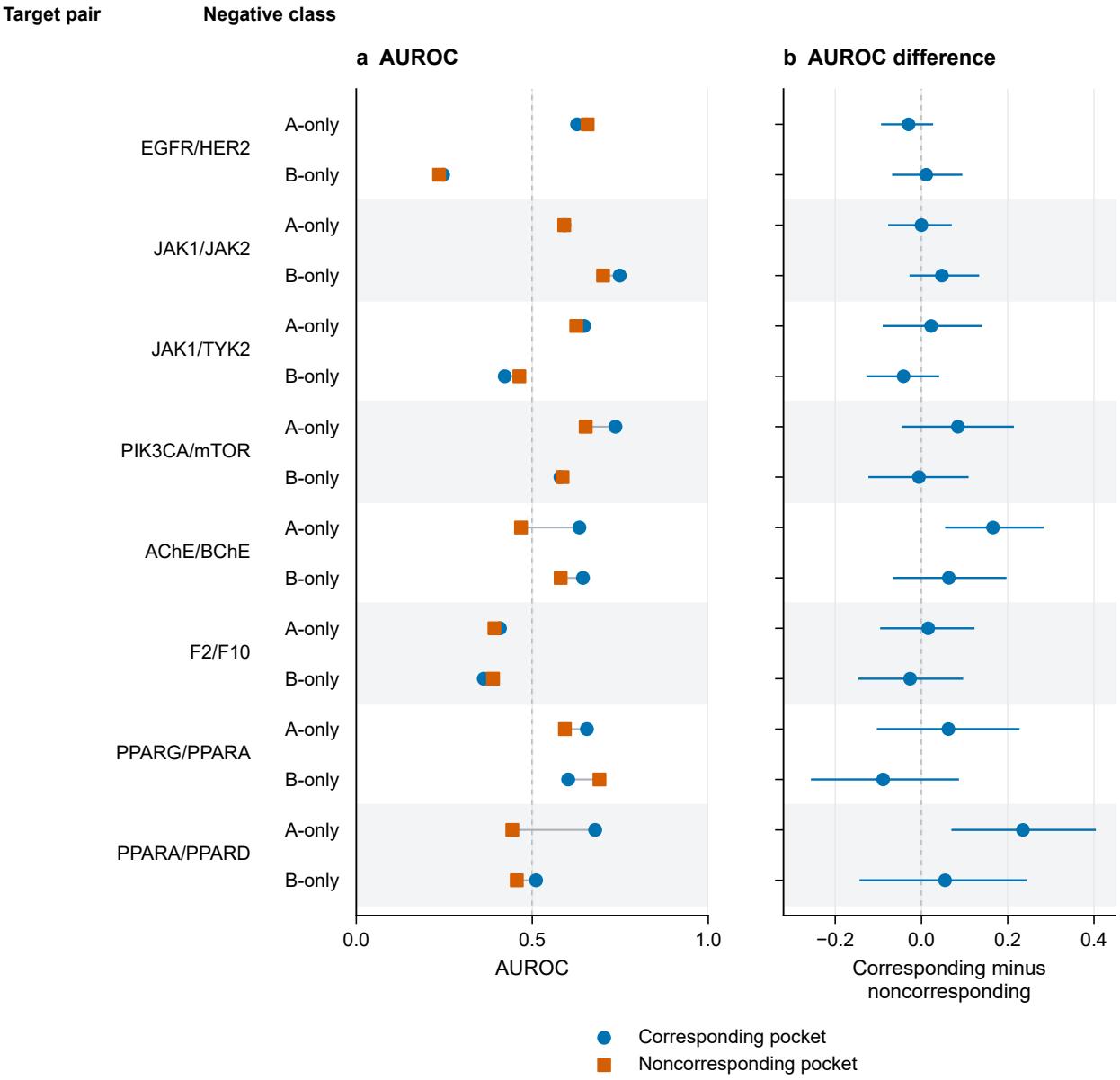


Fig. S8 Full pocket-control estimates and paired intervals underlying Figure 6. Corresponding and noncorresponding pocket controls. (a) AUROC on the same members using corresponding and noncorresponding target scores. The corresponding target is B for dual versus A-only and A for dual versus B-only. (b) Corresponding minus noncorresponding AUROC with 95% paired bootstrap confidence intervals. All scores are Vina S1. Dual is the positive class in every row

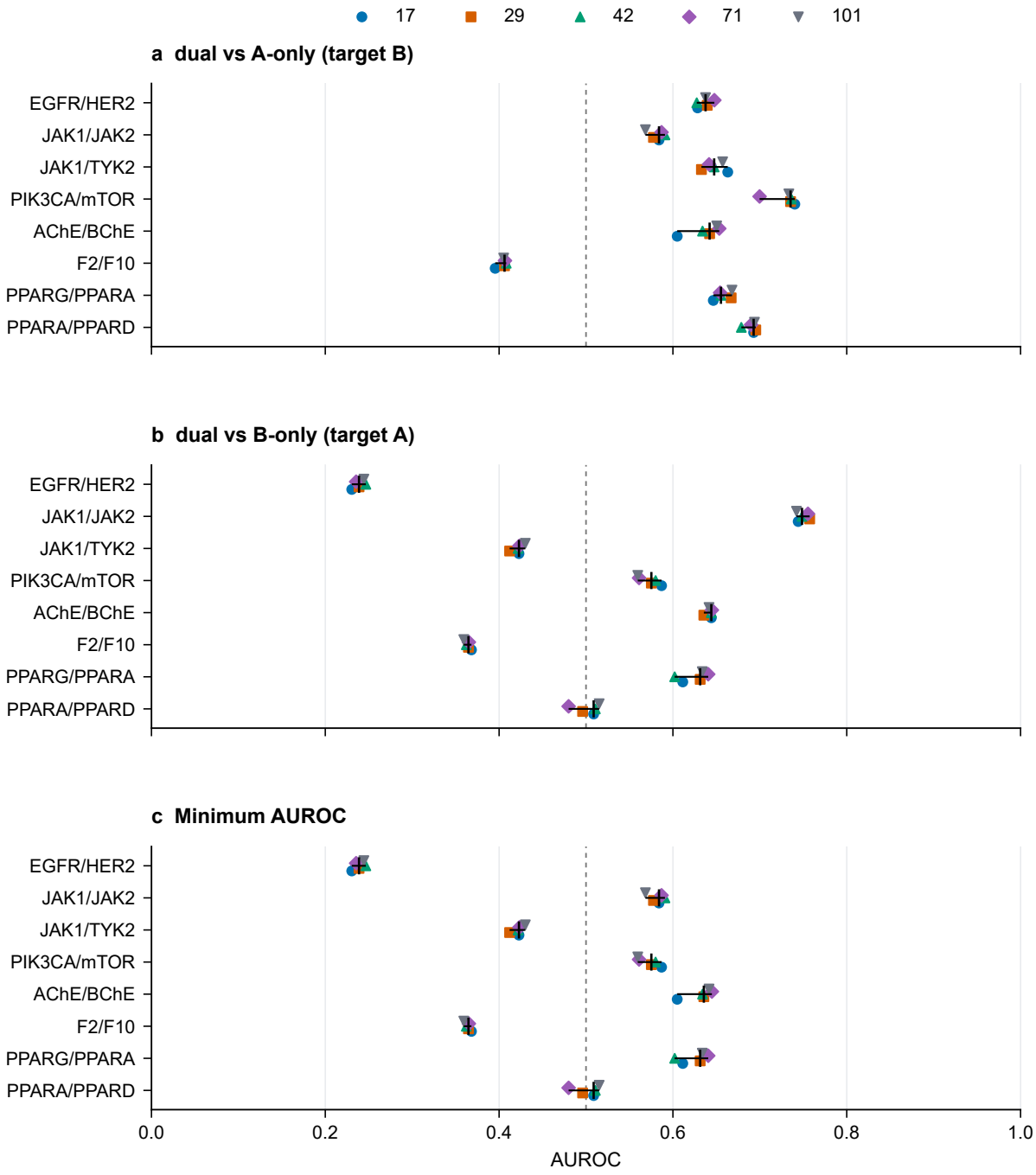


Fig. S9 Full seed sensitivity for directional AUROC and minimum AUROC. Each panel shows five seed-specific estimates, minimum-to-maximum ranges and medians. Range segments are not confidence intervals. Effective populations follow each seed's valid scores; seed 17 loses one A-only AChE/BChE observation relative to seed 42

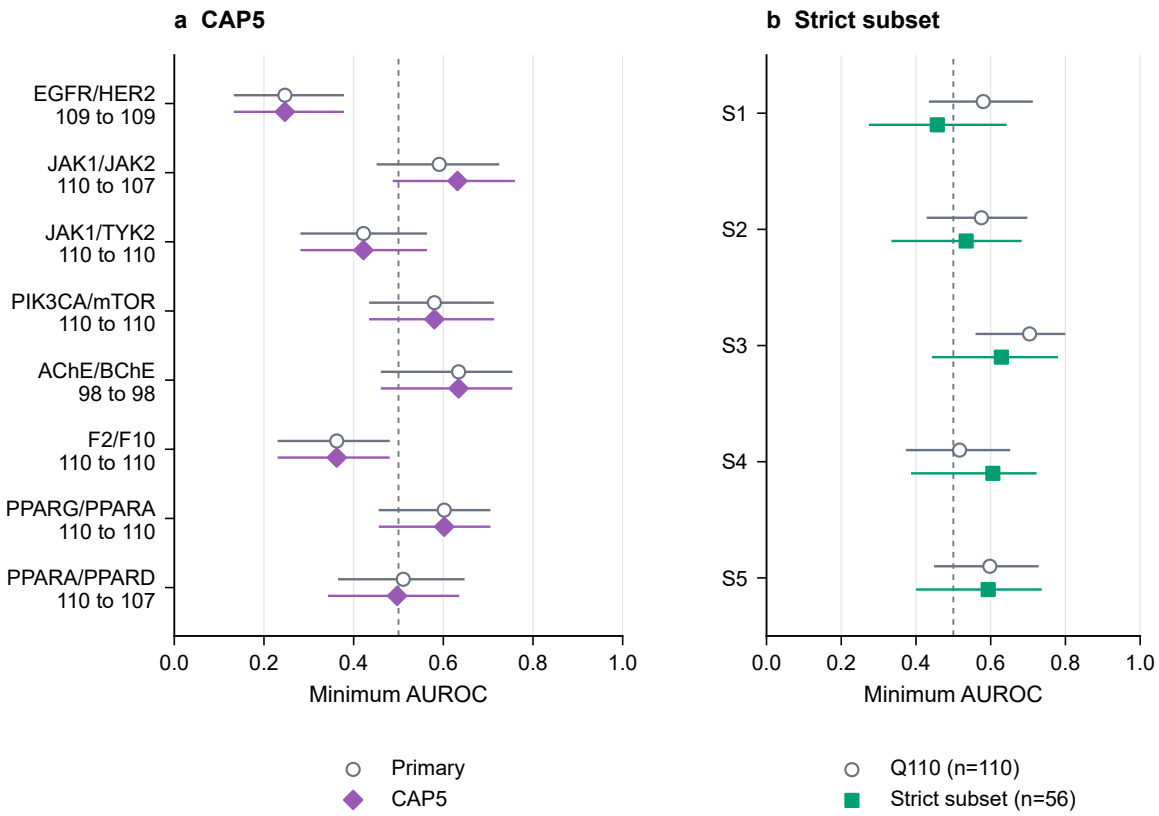
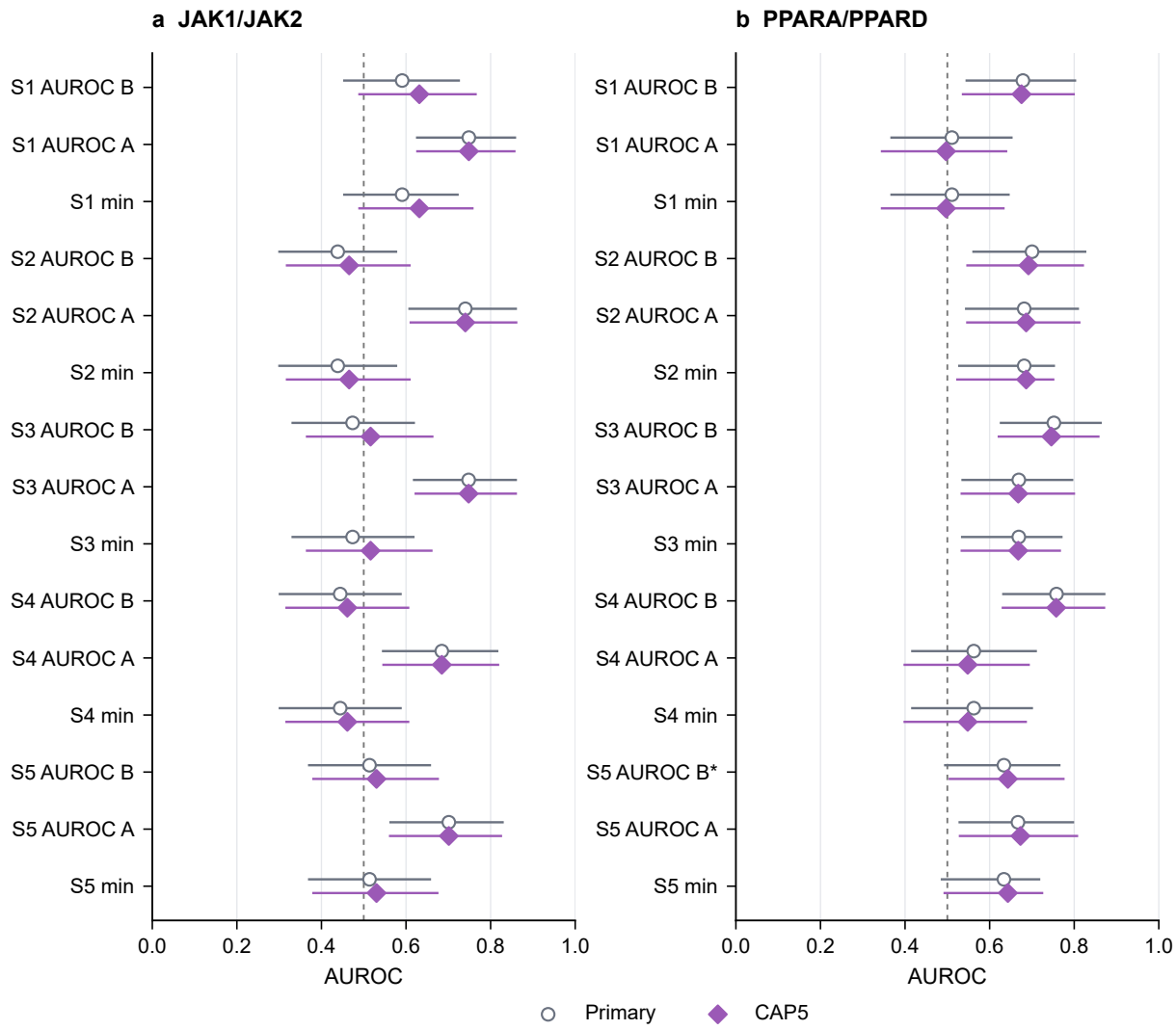


Fig. S10 Sensitivity to evaluation-set membership. (a) Vina minimum AUROC for the primary and CAP5 populations; labels give original and retained pair sizes. CAP5 retains 861 of 867 observations. (b) Minimum AUROC for the five scoring schemes in the 110-member PIK3CA/mTOR set and its 56-member strict-activity subset. Population-size labels include all four experimental classes; directional sample sizes are reported in Tables S10A and S8C. Bars are 95% bootstrap intervals for each population; no paired interval for the between-population change is shown. Full directional CAP5 comparisons, including the PPARA/PPARD S5 dual-versus-A-only interval-boundary change, are shown in Figure S11. Chemistry sensitivity is tabulated in Table S8D



* PPARA/PPARD S5 AUROC B: lower bound of the 95% CI changes from 0.4922 to 0.5030

Fig. S11 Full directional CAP5 sensitivity in the two pairs with excluded members. Each scheme has AUROC B (dual versus A-only), AUROC A (dual versus B-only) and minimum AUROC. Primary and CAP5 estimates retain their respective 95% intervals; no interval for the difference is added. The asterisk identifies PPARA/PPARD S5 AUROC B, whose interval changes from [0.4921875, 0.767578125] to [0.5030241935, 0.7772177419]. This boundary change is reported explicitly without a multiplicity-adjusted significance claim. The other six pairs have unchanged members

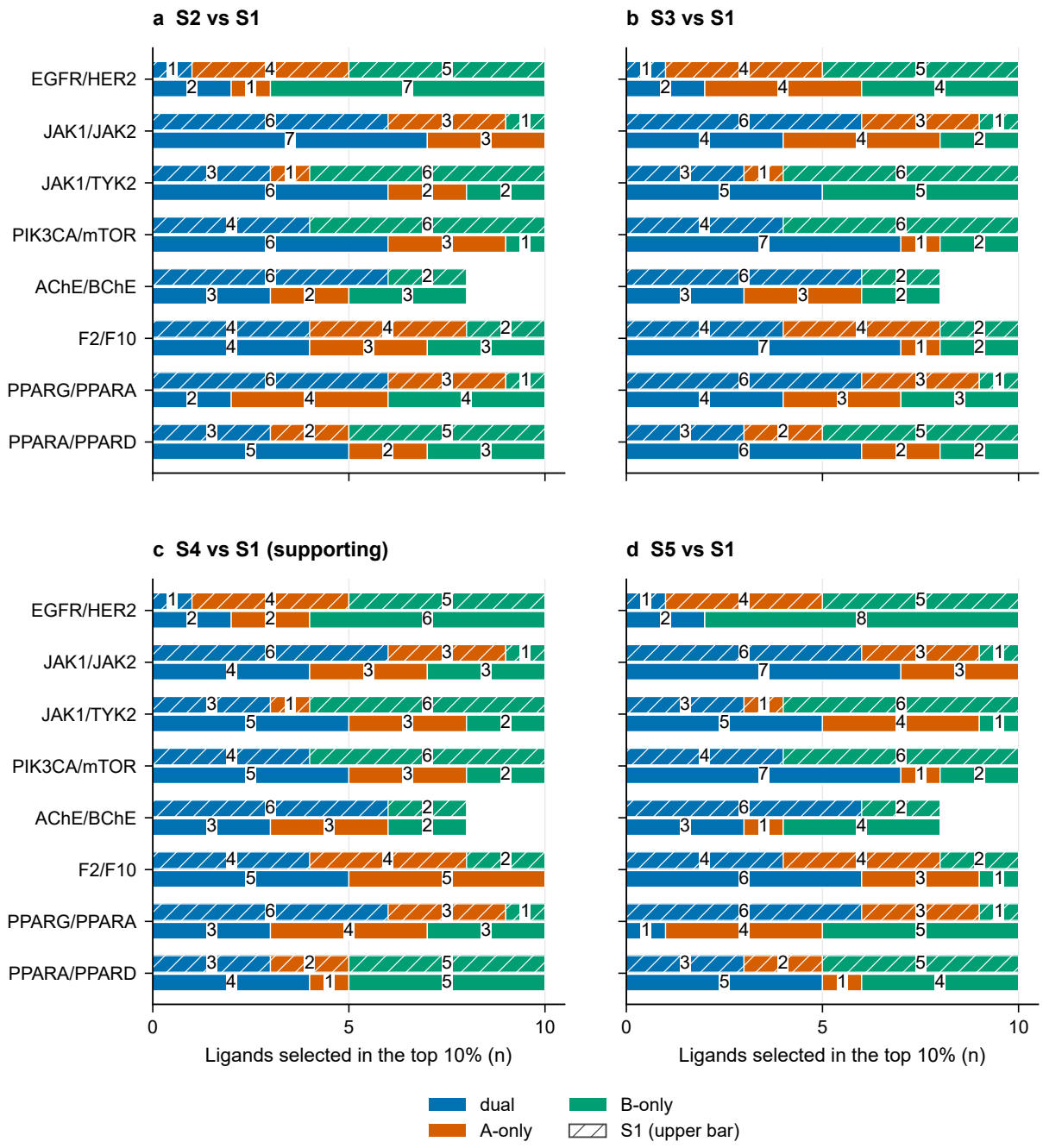
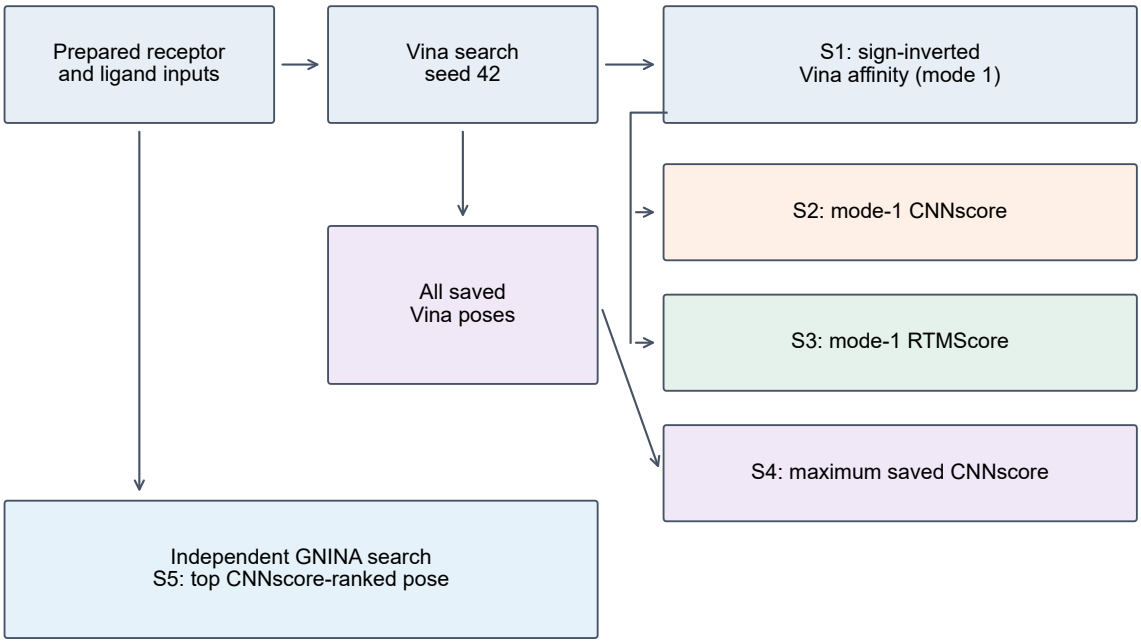


Fig. S12 Class composition of matched top-10% joint-ranking lists. Each target pair has an upper hatched S1 bar and a lower solid new-scheme bar, evaluated on that comparison's common two-target complete-case members. Segment labels give counts. Population sizes and k may differ between method comparisons; the same comparison has identical N and k for its two bars. S4-S1 is supporting. No method-specific S1 list is substituted for these paired baselines



Scheme S1 Scoring workflows S1-S5. Prepared receptor and ligand inputs feed either Vina search (seed 42) or the independent GNINA search. S1 uses sign-inverted Vina affinity from mode 1. S2 and S3 rescore that same mode-1 pose with CNNscore and RTMScore. S4 selects the highest CNNscore among all saved Vina poses. S5 uses independent GNINA docking and selects the top CNNscore-ranked pose. S4-S2 tests saved-pose reranking, whereas S5-S1 compares complete docking workflows